

Requirement Analysis Document

Calculator Web Application

Project Team

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1 1. Introduction

1.1 1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the "Calculator Web Application." This application aims to provide users with a convenient, browser-based tool for performing basic arithmetic calculations.

1.2 1.2 Scope

The project involves developing a client-side web application using standard web technologies (HTML, CSS, and JavaScript).

- **In Scope:** Basic arithmetic operations (Addition, Subtraction, Multiplication, Division), clear functions, and a responsive user interface.
- **Out of Scope:** Scientific functions (trigonometry, logarithms), graphing capabilities, history logging databases, or user authentication.

1.3 1.3 Definitions

- **UI (User Interface):** The visual part of the application where the user interacts.
- **Client-side:** Operations performed locally on the user's device/browser without server intervention.
- **Operand:** Numbers used in the calculation.
- **Operator:** Symbols representing mathematical actions (+, -, *, /).

2 2. Stakeholders

- **End Users:** Individuals requiring a quick and accessible tool for basic math.
- **Developers:** The technical team responsible for coding the HTML structure, CSS styling, and JavaScript logic.
- **QA Team:** Responsible for verifying that calculations are accurate and the UI is responsive.

3 3. Functional Requirements (FR)

This section outlines the specific behaviors and functions the system must support.

3.1 3.1 User Input

FR-1.1: The system shall provide clickable buttons for numerical digits (0-9) and a decimal point (.).

FR-1.2: The system shall provide clickable buttons for standard arithmetic operators: Addition (+), Subtraction (-), Multiplication (*), and Division (/).

3.2 3.2 Operations Logic

FR-2.1: The system shall correctly calculate the result of an operation when the "Equal" (=) button is pressed.

FR-2.2: The system shall support the chaining of operations (e.g., $5 + 5 - 2$) in a logical sequence.

FR-2.3: The system shall respect standard order of operations (PEMDAS) where applicable, or process sequentially based on the simplified scope.

3.3 3.3 Display and Output

FR-3.1: The system shall include a display screen that shows the current input (operands) as they are typed.

FR-3.2: The system shall display the calculated result clearly upon completion of the operation.

3.4 3.4 System Controls

FR-4.1: The system shall provide a "Clear" (C) or "All Clear" (AC) button to reset the current input and result display to zero.

FR-4.2: The system shall provide a "Delete" or "Backspace" button to remove the last entered digit.

3.5 3.5 Error Handling

FR-5.1: The system shall handle division by zero gracefully, displaying a user-friendly error message (e.g., "Error" or "NaN") rather than crashing.

4 4. Non-Functional Requirements (NFR)

4.1 4.1 Usability

NFR-1: The interface shall be intuitive, mimicking the layout of a standard physical calculator.

NFR-2: Buttons shall have sufficient spacing and size to be easily clickable on both desktop mice and mobile touchscreens.

4.2 4.2 Performance

NFR-3: Calculations shall be performed instantly (under 100ms) as they are executed client-side.

4.3 4.3 Compatibility

NFR-4: The application must be responsive and function correctly on all modern web browsers (Chrome, Firefox, Safari, Edge) and device types (Desktop, Tablet, Mobile).

5 5. Assumptions and Constraints

5.1 5.1 Assumptions

- Users have a modern web browser with JavaScript enabled.
- The application does not require offline storage persistence (local storage) for this version.

5.2 5.2 Constraints

- **Technical Constraint:** The project must be built using only HTML5, CSS3, and Vanilla JavaScript. No external frameworks (React, Angular) are to be used for this version.
- **Deployment:** The application will be deployed as a static site.

6 6. Sign-off

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Date: November 20, 2025

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Date: November 20, 2025